RAJALAKSHMI ENGINEERING COLLEGE

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P2

Started on	Wednesday, 19 August 2026, 8:49 AM
State	Finished
Completed on	Wednesday, 19 August 2026, 8:58 AM
Time taken	8 mins 54 secs
Marks	1.00/1.00
Grade	**10.00** out of 10.00 (**100**%)

Question 1 | Correct Mark 1.00 out of 1.00

Given an array `nums` of size `n`, return *the majority element*.

The majority element is the element that appears more than `⌊n / 2⌋` times. You may assume that the majority element always exists in the array.

Example 1:

```
Input: nums = [3,2,3]
Output: 3
```

Example 2:

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

Constraints:

- `n == nums.length`
- `1 <= n <= 5 * 10^4`
- `-2^31 <= nums[i] <= 2^31 - 1`

For example:

Input	Result
3 3 2 3	3
7 2 2 1 1 1 2 2	2

Answer: (penalty regime: 0 %)

```c
#include <stdio.h>

int find(int arr[], int low, int high) {
    if (low == high) {
        return arr[low];
    }
    int mid = low + (high - low) / 2;
    int left = find(arr, low, mid);
    int right = find(arr, mid + 1, high);
    if (left == right) {
        return left;
    }
    int lcount = 0, rcount = 0;
    for (int i = low; i <= high; i++) {
        if (arr[i] == left) lcount++;
        if (arr[i] == right) rcount++;
    }
    return (lcount > rcount) ? left : right;
}

int main() {
    int n;
    scanf("%d", &n);
    int arr[n];
    for (int i = 0; i < n; i++) {
        scanf("%d", &arr[i]);
```

```
    }
    printf("%d\n", find(arr, 0, n - 1));
    return 0;
}
```

	Input	Expected	Got	
✔	3 3 2 3	3	3	✔

Passed all tests! ✔

Correct

Marks for this submission: 1.00/1.00.

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